

Notes: Jiang, Kelly & Xiu (Journal of Finance, 2023)

(Re-)Imag(in)ing Price Trends

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	2
3.1	U.S. stock data and sample construction	2
3.2	Image construction	2
3.3	Labels, training, and benchmarks	3
4	Models and empirical specifications	3
4.1	CNN classification objective	3
4.2	Portfolio turnover	3
4.3	Regression diagnostics	3
5	Identification and empirical validation strategy	3
6	Tables and figures	4
6.1	Figure 1: Tesla OHLC chart from Yahoo! Finance and Figure 2: Generated OHLC images with volume bar and moving average line	4
6.2	Figure 3: Diagram of CNN models and Figure 4: Convolutional filters that detect the next day's price change	4
6.3	Table I: Short-Horizon (One-Week) Portfolio Performance	5
6.4	Figure 5: Cumulative volatility-adjusted returns and Figure 6: Prediction accuracy by decile	5
6.5	Tables II–IV: Longer horizons, performance breakdown, and largest stocks	6
6.6	Tables V–VII: CNN predictions, stock characteristics, and future returns	7
6.7	Tables VIII–IX and Figures 7–8: Image scaling, logistic approximations, CNN1D, and technical indicators	7
6.8	Table X and Figure 9: International transfer and H-L decile portfolio Sharpe ratios	8
6.9	Tables XI–XII: Time-scale transfer	9
7	Interpretive synthesis	9
8	Short summary	10
9	Conclusion	10
9.1	Core conclusion	10
9.2	Economic intuition	10
9.3	Takeaway	10

1 Big picture

- **Question.** Do stock price charts contain complex return-predictive patterns beyond standard hand-crafted trend signals such as momentum, reversal, and moving-average rules?
- **Core idea.** The paper represents historical prices as images and lets a convolutional neural network learn the shapes that best predict future returns.
- **Input data.** Each image is an OHLC-style chart of adjusted stock prices, optionally with volume bars and a moving-average line, over 5, 20, or 60 trading days.
- **Prediction target.** The CNN is trained as a classifier: the label equals one if the subsequent return over 5, 20, or 60 days is positive and zero otherwise.
- **Empirical punchline.** CNN image signals produce economically large out-of-sample decile-spread returns and Sharpe ratios, outperforming familiar price-trend benchmarks.
- **Conceptual message.** The visual representation matters because it allows the model to use direction, volatility, moving averages, and volume jointly.

2 Incremental contribution of the paper

- **Methodological contribution.** The paper brings computer-vision methods into empirical asset pricing by turning price histories into images and using CNNs to discover predictive price patterns.
- **Measurement contribution.** It constructs standardized OHLC image objects from CRSP daily open, high, low, close, return, and volume data.
- **Empirical contribution.** It documents that image-based signals generate stronger out-of-sample portfolio performance than momentum, short-term reversal, weak reversal, and moving-average trend indicators.
- **Interpretability contribution.** It compares CNN signals to standard stock characteristics, logistic approximations, one-dimensional CNNs, and thousands of technical indicators to understand what the model learns.
- **External-validity contribution.** International transfer and time-scale transfer show that patterns learned in one context can help forecast returns in other markets and frequencies.
- **Theory contribution.** The evidence motivates new models of price dynamics that are nonlinear, visual, and potentially self-similar across time scales.

3 Data and measurement

3.1 U.S. stock data and sample construction

Daily stock data come from CRSP for NYSE, AMEX, and NASDAQ stocks. The sample covers 1993–2019 because daily opening, high, and low prices become available in June 1992. The test period is 2001–2019. Training and validation use 1993–2000. The design fixes the CNN after the early training window, so later performance is a long out-of-sample experiment.

Interpretation: This design emphasizes genuine forecasting rather than repeated in-sample tuning.

3.2 Image construction

The adjusted close price sequence is constructed from returns:

$$p_{t+1} = (1 + RET_{t+1})p_t.$$

The first close in the image is normalized to one. Opening, high, and low levels are scaled relative to closing prices. The main image lengths are 5, 20, and 60 trading days.

Interpretation: This construction forces the model to learn relative shapes and dynamics, not raw price levels or split/dividend artifacts.

3.3 Labels, training, and benchmarks

Each image receives label $y = 1$ if the future return over the target horizon is positive. The horizon can be 5, 20, or 60 days. The paper combines three image windows with three label horizons, producing nine baseline models. Benchmarks include MOM, STR, WSTR, and TREND.

Interpretation: The notation Ix/Ry separates the information window from the return horizon.

4 Models and empirical specifications

4.1 CNN classification objective

$$L(y, \hat{y}) = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y}).$$

Interpretation: The CNN outputs a probability that the subsequent return is positive. Cross-entropy loss penalizes probability forecasts that disagree with the realized sign.

4.2 Portfolio turnover

$$Turnover = \frac{1}{MT} \sum_t \sum_i \left| w_{i,t+1} - \frac{w_{i,t}(1 + r_{i,t+1})}{1 + \sum_j w_{j,t} r_{j,t+1}} \right|.$$

Interpretation: Turnover matters because short-horizon image strategies can be profitable in gross returns but costly to trade.

4.3 Regression diagnostics

The paper regresses CNN forecasts on stock characteristics and technical signals, and then predicts future return signs using both CNN forecasts and controls.

Interpretation: These regressions test whether CNN forecasts are just repackaged momentum, reversal, size, liquidity, or volatility signals.

5 Identification and empirical validation strategy

The paper is a predictive validation study rather than a causal design. Credibility comes from: (i) a fixed early training window; (ii) a long 2001–2019 test period; (iii) portfolio sorts formed only with information available before the holding period; (iv) comparisons with standard benchmarks; (v) value-weight and large-stock tests; (vi) diagnostic controls; (vii) comparison with thousands of technical rules; and (viii) international and time-scale transfer.

Interpretation: The main threat is overfitting. The response is out-of-sample persistence plus external validation.

6 Tables and figures

6.1 Figure 1: Tesla OHLC chart from Yahoo! Finance and Figure 2: Generated OHLC images with volume bar and moving average line

Read: Figure 1 shows the familiar visual object motivating the paper. Figure 2 shows the standardized machine-learning version.

Interpretation: The key methodological step is converting market history into a visual field that a CNN can process.

Economic message: Representation is part of the empirical model.



Figure 1. Tesla OHLC chart from Yahoo! Finance. This figure displays an OHLC chart for Tesla stock with 20-day moving average price line and daily volume bars. Data are daily from January 1, 2020, to August 18, 2020. (Color figure can be viewed at wileyonlinelibrary.com)

and quarterly price trajectories, respectively). The width of an n -day image is therefore $3n$ pixels. We replace prices by CRSP-adjusted returns to translate the opening closing high and low prices into relative scales that abstract from

(a) Figure 1: Tesla OHLC chart

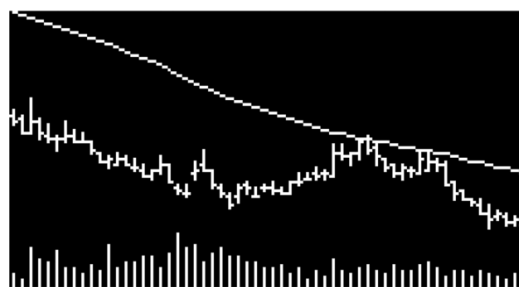


Figure 2. Generated OHLC images with volume bar and moving average line. The figure displays market data image for 60 days of data.

II. The CNN Model

(b) Figure 2: generated OHLC image

6.2 Figure 3: Diagram of CNN models and Figure 4: Convolutional filters that detect the next day's price change

Read: Figure 3 shows architecture by image length. Figure 4 shows how convolutional filters detect local nonlinear price configurations.

Interpretation: The CNN learns local visual filters rather than researcher-defined trend variables.

Economic message: The model automates technical feature extraction.

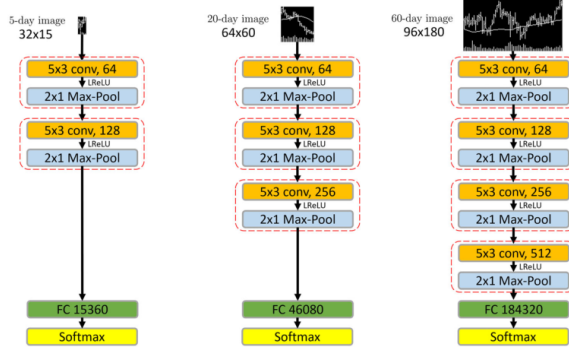


Figure 3. Diagram of CNN models. This figure displays diagrams of a five-day (left), 20-day (middle), and 60-day (right) CNN model. The 5/20/60-day CNN models are built with 2/3/4 CNN building blocks as described in Figure A2. The notation $H \times W$ shows the output size of the CNN building block, where H is the height and W the width. The number following "conv" or "Max-Pool" is the depth (number of channels, for example, 64, 128, 256, and 512). The output of the last CNN block is flattened to a vector and fed to a fully connected (FC) layer, where the reported input size (after FC) is calculated as the product $H \times W \times D$ from the last CNN building block. The final Softmax layer yields the probabilities of "up" and "down." (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 3: CNN architecture

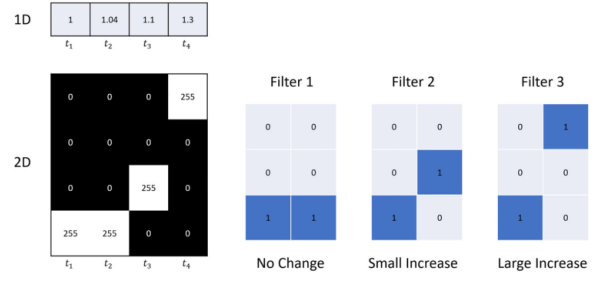


Figure 4. Convolutional filters that detect the next day's price change. The "1D" figure on the left displays a snippet of time-series prices. The "2D" figure on the left translates this time-series into a black and white image where "255" represents a white pixel (corresponding to a price entry) and "0" represents empty space in the image. The image is a discretized representation of the prices. For illustration, each unit on the y-axis of the image is 0.1. Because the difference between 1.04 and 1.0 is smaller than 0.05, the discretized prices on the image are flat over the first two periods. The figure on the right displays three 3×2 convolutional filters that detect no change, a small increase, and a large increase in price, respectively. (Color figure can be viewed at wileyonlinelibrary.com)

instead extracts predictive patterns from the market data series within the CNN itself.

In short, ingesting data in the form of an image allows the model to isolate

(b) Figure 4: filters

6.3 Table I: Short-Horizon (One-Week) Portfolio Performance

Read: Equal-weight CNN strategies produce very large one-week high-minus-low Sharpe ratios. I5/R5, I20/R5, and I60/R5 H-L Sharpe ratios are about 7.15, 6.75, and 4.89. Value-weight performance is lower but still meaningful.

Interpretation: This is the paper's main proof-of-concept: CNN scores rank stocks monotonically by future returns.

Economic message: Chart images contain return-predictive information that standard trend variables leave unused.

Table I
Short-Horizon (One-Week) Portfolio Performance

This table reports results on the performance of equal-weight (top panel) and value-weight (bottom panel) decile portfolios sorted on out-of-sample predicted "up" probability. Each panel reports the average holding-period return and annualized Sharpe ratio. ***, **, and * indicate significant at the 1%, 5%, and 10% significance level, respectively. We also report the monthly turnover of each strategy.

	Equal-Weight													
	I5/R5		I20/R5		I60/R5		MOM/R5		STR/R5		WSTR/R5		TREND/R5	
	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR
Low	-0.28	-1.92	-0.32	-1.94	-0.21	-1.10	0.15	0.44	-0.01	-0.03	-0.08	-0.34	-0.11	-0.46
2	-0.04	-0.27	-0.04	-0.21	0.02	0.12	0.10	0.44	0.06	0.35	0.04	0.24	0.01	0.05
3	0.03	0.15	0.04	0.20	0.07	0.35	0.10	0.50	0.09	0.58	0.08	0.48	0.05	0.30
4	0.08	0.41	0.08	0.43	0.11	0.58	0.10	0.57	0.10	0.67	0.09	0.58	0.08	0.50
5	0.09	0.48	0.12	0.65	0.14	0.75	0.10	0.63	0.11	0.68	0.09	0.60	0.10	0.64
6	0.14	0.70	0.15	0.80	0.16	0.88	0.12	0.76	0.11	0.70	0.11	0.65	0.11	0.71
7	0.17	0.84	0.19	0.97	0.17	0.93	0.13	0.83	0.11	0.64	0.13	0.75	0.13	0.78
8	0.22	1.06	0.23	1.19	0.20	1.08	0.14	0.90	0.12	0.62	0.14	0.72	0.16	0.85
9	0.30	1.48	0.27	1.40	0.22	1.23	0.15	0.91	0.16	0.68	0.18	0.81	0.23	1.04
High	0.54	2.89	0.52	2.76	0.33	1.85	0.16	0.78	0.38	1.19	0.46	1.56	0.48	1.58
H-L	0.83***	7.15	0.84***	6.75	0.54***	4.89	0.02	0.07	0.39***	1.76	0.53***	2.84	0.59***	2.92
Turnover	690%		667%		619%		123%		341%		660%		499%	

(Continued)

Table I, part 1

Table I—Continued

Value-Weight

	I5/R5		I20/R5		I60/R5		MOM/R5		STR/R5		WSTR/R5		TREND/R5	
	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR
	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR	Ret	SR
Low	-0.03	-0.19	-0.03	-0.16	-0.02	-0.12	0.03	0.06	0.04	0.16	-0.02	-0.09	0.02	0.08
2	0.00	0.00	0.02	0.12	0.01	0.06	0.02	0.05	0.04	0.22	0.03	0.14	0.02	0.10
3	0.04	0.24	0.04	0.24	0.04	0.24	0.07	0.27	0.06	0.34	0.06	0.34	0.06	0.35
4	0.06	0.35	0.06	0.32	0.06	0.31	0.08	0.36	0.09	0.54	0.06	0.36	0.07	0.41
5	0.06	0.34	0.06	0.34	0.06	0.34	0.08	0.41	0.09	0.57	0.08	0.48	0.08	0.50
6	0.09	0.50	0.09	0.51	0.06	0.32	0.08	0.46	0.10	0.57	0.10	0.60	0.09	0.55
7	0.11	0.58	0.09	0.49	0.08	0.45	0.08	0.51	0.10	0.51	0.11	0.63	0.12	0.66
8	0.12	0.62	0.11	0.61	0.09	0.49	0.10	0.62	0.13	0.63	0.15	0.73	0.14	0.71
9	0.16	0.81	0.11	0.59	0.11	0.61	0.11	0.62	0.13	0.81	0.18	0.72	0.14	0.57
High	0.20	0.91	0.20	0.99	0.14	0.77	0.12	0.63	0.16	0.46	0.18	0.59	0.18	0.55
H-L	0.23***	1.49	0.22***	1.74	0.16***	1.44	0.12	0.33	0.13*	0.44	0.21***	0.77	0.18***	0.68
Turnover	758%		728%		671%		169%		433%		733%		575%	

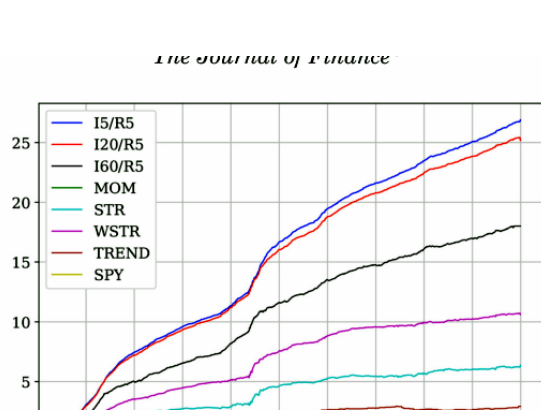
Table I, part 2

6.4 Figure 5: Cumulative volatility-adjusted returns and Figure 6: Prediction accuracy by decile

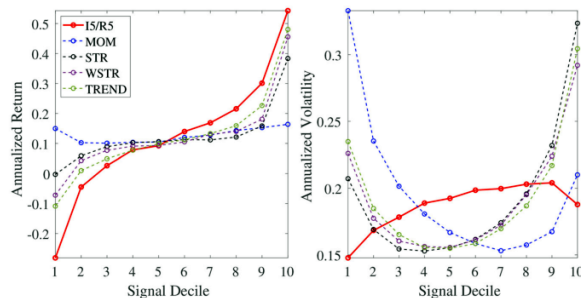
Read: Figure 5 shows cumulative volatility-scaled returns; Figure 6 shows monotone realized returns across CNN forecast deciles.

Interpretation: The signal is persistent over time and calibrated across the forecast distribution.

Economic message: A good prediction model should rank assets, not merely identify isolated winners.



(a) Figure 5



ure 6. Prediction accuracy by decile. The left panel plots average realized returns: a decile of I5/R5 CNN model forecasts and for each decile of the benchmark signals. Returnages are based on cross-sectional decile averages in each period that are then averaged out. The right panel plots the time-series volatility of decile returns. (Color figure can be viewed at onlinelibrary.com)

6.5 Tables II–IV: Longer horizons, performance breakdown, and largest stocks

Table II
Performance of Monthly and Quarterly Strategies

This table reports results on the performance of equal-weight decile portfolios sorted on out-of-sample predicted "up" probability. Each panel reports the average holding-period return and standard deviation (SD) for each decile portfolio. Each panel also reports the average holding-period return and standard deviation (SD) for each decile portfolio. Each panel also reports the average holding-period return and standard deviation (SD) for each decile portfolio. Each panel also reports the average holding-period return and standard deviation (SD) for each decile portfolio.

	I5/R5		I20/R5		I60/R5		MOM		STR		WSTR		TREND	
	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD
Low	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00
2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
4	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
8	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
9	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
High	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
High	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Treasury	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%

Table III—Continued

	I5/R5		I20/R5		I60/R5		MOM		STR		WSTR		TREND	
	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD
Low	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
4	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
8	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
9	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
High	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
High	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Treasury	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%

Table IV
Equal-Weight Portfolio Performance Breakdown

This table provides breakdowns of performance of equal-weight decile portfolios sorted on out-of-sample predicted "up" probability. Each panel reports the average holding-period return and standard deviation (SD) for each decile portfolio. Each panel also reports the average holding-period return and standard deviation (SD) for each decile portfolio. Each panel also reports the average holding-period return and standard deviation (SD) for each decile portfolio. Each panel also reports the average holding-period return and standard deviation (SD) for each decile portfolio.

	I5/R5		I20/R5		I60/R5		MOM		STR		WSTR		TREND	
	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD
Low	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00
2	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00
3	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00
4	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00
5	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00
6	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00
7	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00
8	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00
9	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00
High	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00
High	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00	-0.00	0.00
Treasury	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%

Read: Table II shows monthly and quarterly performance. Table III decomposes monthly returns into days 1–5 and days 6–20. Table IV restricts the sample to the largest 500 stocks.

Interpretation: These tests reduce concerns that the signal is only a high-turnover or tiny-stock phenomenon.

Economic message: Image predictability remains economically relevant under more implementable portfolio restrictions.

Table III—Continued

	I5/R5		I20/R5		I60/R5		MOM		STR		WSTR		TREND	
	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD
Low	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
4	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
8	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
9	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
High	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
High	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Treasury	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%

Table IV—Continued

	I5/R5		I20/R5		I60/R5		MOM		STR		WSTR		TREND	
	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD
Low	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
4	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
8	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
9	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
High	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
High	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Treasury	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%

Table IV—Continued															
	Value Weight														
	I5/R5		I20/R5		I60/R5		MOM/R5		STR/R5		WSTR/R5		TREND/R5		
	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD	Ret	SD	
Low	-0.01	-0.01	0.00	0.18	0.02	0.18	0.00	0.20	0.00	0.13	0.00	0.20	0.08	0.17	0.09
2	-0.01	0.01	0.01	0.13	0.02	0.13	0.01	0.17	0.01	0.13	0.01	0.17	0.08	0.17	0.09
3	0.00	0.03	0.00	0.11	0.05	0.24	0.00	0.17	0.01	0.10	0.00	0.16	0.04	0.07	0.07
4	0.00	0.03	0.00	0.11	0.05	0.24	0.00	0.17	0.01	0.10	0.00	0.16	0.04	0.07	0.07
5	0.00	0.03	0.00	0.11	0.05	0.24	0.00	0.17	0.01	0.10	0.00	0.16	0.04	0.07	0.07
6	0.04	0.02	0.02	0.07	0.00	0.05	0.04	0.08	0.02	0.11	0.07	0.07	0.07	0.11	0.07
7	0.09	0.02	0.07	0.04	0.00	0.05	0.04	0.08	0.02	0.11	0.07	0.07	0.07	0.11	0.07
8	0.07	0.02	0.07	0.04	0.00	0.05	0.04	0.08	0.02	0.11	0.07	0.07	0.07	0.11	0.07
9	0.07	0.02	0.07	0.04	0.00	0.05	0.04	0.08	0.02	0.11	0.07	0.07	0.07	0.11	0.07
10	0.07	0.02	0.07	0.04	0.00	0.05	0.04	0.08	0.02	0.11	0.07	0.07	0.07	0.11	0.07
11	0.07	0.02	0.07	0.04	0.00	0.05	0.04	0.08	0.02	0.11	0.07	0.07	0.07	0.11	0.07
12	0.07	0.02	0.07	0.04	0.00	0.05	0.04	0.08	0.02	0.11	0.07	0.07	0.07	0.11	0.07
13	0.13	0.08	0.12	0.03	0.13	0.07	0.05	0.04	0.04	0.02	0.07	0.20	0.08	0.05	0.04
14	0.13	0.08	0.12	0.03	0.13	0.07	0.05	0.04	0.04	0.02	0.07	0.20	0.08	0.05	0.04
15	0.13	0.08	0.12	0.03	0.13	0.07	0.05	0.04	0.04	0.02	0.07	0.20	0.08	0.05	0.04
High	0.17	0.12	0.09	0.11	0.06	0.12	0.06	0.12	0.02	0.12	0.06	0.12	0.09	0.11	0.06
2	0.17	0.12	0.09	0.12	0.06	0.12	0.06	0.12	0.02	0.12	0.06	0.12	0.09	0.11	0.06
3	0.17	0.12	0.09	0.12	0.06	0.12	0.06	0.12	0.02	0.12	0.06	0.12	0.09	0.11	0.06
4	0.17	0.12	0.09	0.12	0.06	0.12	0.06	0.12	0.02	0.12	0.06	0.12	0.09	0.11	0.06
5	0.17	0.12	0.09	0.12	0.06	0.12	0.06	0.12	0.02	0.12	0.06	0.12	0.09	0.11	0.06
6	0.17	0.12	0.09	0.12	0.06	0.12	0.06	0.12	0.02	0.12	0.06	0.12	0.09	0.11	0.06
7	0.17	0.12	0.09	0.12	0.06	0.12	0.06	0.12	0.02	0.12	0.06	0.12	0.09	0.11	0.06
8	0.17	0.12	0.09	0.12	0.06	0.12	0.06	0.12	0.02	0.12	0.06	0.12	0.09	0.11	0.06
9	0.17	0.12	0.09	0.12	0.06	0.12	0.06	0.12	0.02	0.12	0.06	0.12	0.09	0.11	0.06
10	0.17	0.12	0.09	0.12	0.06	0.12	0.06	0.12	0.02	0.12	0.06	0.12	0.09	0.11	0.06
11	0.17	0.12	0.09	0.12	0.06	0.12	0.06	0.12	0.02	0.12	0.06	0.12	0.09	0.11	0.06
12	0.17	0.12	0.09	0.12	0.06	0.12	0.06	0.12	0.02	0.12	0.06	0.12	0.09	0.11	0.06
13	0.17	0.12	0.09	0.12	0.06	0.12	0.06	0.12	0.02	0.12	0.06	0.12	0.09	0.11	0.06
14	0.17	0.12	0.09	0.12	0.06	0.12	0.06	0.12	0.02	0.12	0.06	0.12	0.09	0.11	0.06
15	0.17	0.12	0.09	0.12	0.06	0.12	0.06	0.12	0.02	0.12	0.06	0.12	0.09	0.11	0.06

6.6 Tables V–VII: CNN predictions, stock characteristics, and future returns

Read: These tables compare CNN forecasts to standard stock characteristics and test whether CNN predictions forecast returns after controls.

Interpretation: CNN forecasts overlap with known variables but are not spanned by them.

Economic message: The signal is not simply disguised momentum, reversal, illiquidity, or size.

Table VI
CNN Predictions and Standard Stock Characteristics

This table reports slope coefficients and R^2 s from panel logistic regressions of CNN model forecasts on stock characteristics. Panel regressions are estimated during the test sample using CNN models estimated in the training sample. ***, **, and * indicate significant at the 1%, 5%, and 10% significance level, respectively, using Newey-West standard errors.

	5DIP	20DIP	60DIP
MOM	-0.10***	0.01**	0.40***
STR	-0.00***	0.27***	0.24***
Lag Weekly Return	-0.00***	-1.00***	-0.80***
TREND	0.12***	0.40***	0.23***
Beta	0.11***	0.15***	0.22***
Volatility	-0.00***	-0.20***	-0.24***
SPWH	-0.00***	-0.00***	-0.00***
Bid-Ask	0.10***	-0.11***	-0.08***
Dollar Volume	0.20***	0.16***	-1.22***
Zero Trade	-0.00***	0.00	0.22***
Price Delay	-0.01***	-0.01**	0.00
Size	0.21***	0.40***	0.44***
Illiquidity	0.00***	0.10***	-1.43***
McFadden R^2	8.20	8.56	9.78

Table VII
CNN, Future Returns, and Standard Stock Characteristics

This table reports slope coefficients from panel logistic regressions of future returns on standard stock characteristics. Panel regressions are estimated during the test sample using CNN models estimated in the training sample. ***, **, and * indicate significant at the 1%, 5%, and 10% significance level, respectively, using Newey-West standard errors. The table also reports out-of-sample McFadden R^2 , computed by minimizing over a model estimated on the training sample against that of a benchmark model using the same stock characteristics as the predictor (to ensure high quality estimates of sample sizes for each stock, we require a stock to have at least three years of data; 10 observations for quarterly returns). For stocks that did not meet this criteria, we tested on the overall in-sample mean across all stocks. The same adjustment applies to Table VIII.)

	3MRS	12MRS	36MRS
CNN	0.20***	0.24***	0.30***
MOM	0.00***	0.00***	0.00***
STR	0.00***	0.00***	0.00***
Lag Weekly Return	0.00***	0.00***	0.00***
TREND	0.11***	0.00***	0.12***
Beta	0.12***	0.00***	0.10***
Volatility	-0.00***	-0.00***	-0.00***
SPWH	-0.00***	-0.00***	-0.00***
Bid-Ask	-0.00***	-0.00***	-0.00***
Dollar Volume	-0.10***	-0.10***	-0.10***
Zero Trade	0.00	0.00	0.00
Price Delay	0.00	0.00	0.00
Size	0.21***	0.21***	0.21***
Illiquidity	-0.00***	-0.00***	-0.00***
McFadden R^2	0.10	0.08	0.10

correlates are size, volatility, bid-ask spread, illiquidity, and dollar volume, all of which reach correlations near 30% and associate most strongly with fore-

6.7 Tables VIII–IX and Figures 7–8: Image scaling, logistic approximations, CNN1D, and technical indicators

Read: Tables VIII and IX show that image scaling is crucial and that CNNs outperform logistic and one-dimensional alternatives. Figure 8 compares CNNs with 7,846 technical rules.

Interpretation: Representation, not just model flexibility, drives performance.

Economic message: Learned image patterns dominate a broad universe of manually defined rules.

Table VIII
Logistic Regressions Using Market Data with Image Scaling

This table reports results of logistic regressions on out-of-sample CNN forecasts and realized future return indicators on daily price and volume data. Panel regressions are estimated during the test sample using CNN models estimated in the training sample. * indicates significant at the 1% significance level using Newey-West standard errors. McFadden R^2 s are associated with regressions using test sample data. To facilitate comparison of predictive performance, we report out-of-sample McFadden R^2 s, which compare the cross-entropy loss of a model (estimated in the training sample) against that of a benchmark model using the stock-level in-sample mean as the predictor (as ma lag 1 equals the simple average of close lag 1 to 5 by definition, we remove close lag 5 in the regression to avoid multicollinearity).

	CNN as Dep. Var.			Positive Return Indicator as Dep. Var.							
	15/R5	15/R20	15/R60								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
				2.21			2.51		1.90	2.30	
CNN											
open lag 1	-0.58*	-0.00	0.28*		-0.25	-0.16		-0.13	-0.11		0.12
open lag 2	0.27*	0.54*	0.22*		0.08	0.05		0.13	0.09		-0.10
open lag 3	0.15*	-0.20*	-0.09*		-0.08	-0.10		0.00	0.02		0.14
open lag 4	0.16*	0.01	0.04*		-0.08	-0.09		-0.06	-0.04		0.14
open lag 5	0.00	0.10*	0.37*		0.03	0.01		-0.00	-0.01		-0.25
high lag 1	2.65*	1.50*	1.08*		0.39	0.09		0.17	0.08		0.07
high lag 2	-0.26*	0.47*	0.77*		0.05	0.08		0.05	0.01		-0.12
high lag 3	-0.42*	-0.18*	-0.20*		0.12	0.16		-0.00	0.02		0.06
high lag 4	-0.45*	-0.12*	-0.20*		0.06	0.10		0.23	0.23		0.01
high lag 5	-0.04*	0.41*	-0.31*		0.01	0.03		-0.22	-0.24		0.02
low lag 1	1.90*	0.94*	-0.06*		0.48	0.20		0.40	0.30		0.21
low lag 2	-0.10*	0.77*	0.74*		0.12	0.16		0.07	0.02		0.14
low lag 3	-0.19*	-0.06*	-0.18*		0.05	0.08		-0.11	-0.10		-0.10
low lag 4	0.20*	0.43*	-0.12*		0.03	0.01		0.15	0.12		-0.17
low lag 5	-0.04*	-0.04*	-0.21*		0.09	0.12		-0.21	-0.20		-0.11
close lag 1	-5.50*	-4.06*	-2.30*		-0.71	-0.16		-1.01	-0.73		-0.63
close lag 2	0.54*	0.15*	0.06*		-0.06	-0.08		-0.49	-0.49		-0.04
close lag 3	0.46*	0.62*	0.14*		-0.12	-0.12		-0.36	-0.38		-0.05
close lag 4	0.55*	0.09*	0.06*		-0.19	-0.19		-0.50	-0.49		-0.05

(Continued)

Table VIII—Continued

	CNN as Dep. Var.			Positive Return Indicator as Dep. Var.							
	15/R5	15/R20	15/R60								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
ma lag 1	-0.95*	-0.66*	-0.28*		0.34	0.28		2.50	2.50		1.47
ma lag 2	0.44*	0.50*	0.11*		-0.28	-0.29		-0.34	-0.38		-0.65
ma lag 3	0.03*	-0.69*	0.24*		-0.07	-0.10		-0.56	-0.55		-0.07
ma lag 4	-0.03	0.33*	0.63*		0.49	0.44		0.42	0.46		-0.35
ma lag 5	0.23*	-0.46*	-0.81*		-0.22	-0.19		-0.00	0.01		0.60
vol lag 1	0.81*	0.71*	0.42*		0.06	0.00		0.12	0.08		0.16
vol lag 2	0.21*	0.52*	0.58*		0.05	0.04		-0.01	-0.03		0.03
vol lag 3	0.13*	0.15*	0.37*		0.04	0.03		-0.01	-0.02		0.08
vol lag 4	-0.07*	0.12*	0.09*		0.05	0.05		0.04	0.04		0.18
vol lag 5	0.05*	0.01	0.30*		0.02	0.02		0.05	0.05		0.04
McFadden R^2	35.16	33.15	21.86								
OOS McFadden R^2				0.73	0.55	0.73	0.47	0.47	0.46	1.33	1.39

Table IX
Portfolio Performance of CNN versus Logistic Model and CNN1D

This table reports annualized Sharpe ratios of long-short decile spread portfolios (with equal-weights or value-weights) sorted on out-of-sample predicted "up" probability from each model. "CNN" is the baseline 2D CNN, "logistic" is the simple logistic regression model, and "CNN1D" is the time-series CNN model. "Image scale" indicates the same high/low price normalization used to produce images. "Cum. ret. scale" indicates that market prices are normalized by the closing price at the start of each image. "Devol. ret. scale" indicates that prices each day are converted to returns then normalized by conditional volatility.

	Five-Day Images		20-Day Images		60-Day Images	
	Equal-Weight	Value-Weight	Equal-Weight	Value-Weight	Equal-Weight	Value-Weight
Five-Day Return Horizon						
CNN	7.15	1.49	6.75	1.74	4.89	1.44
Logistic (image scale)	5.56	1.60	4.82	1.52	4.91	1.93
Logistic (cum. ret. scale)	2.50	0.23	1.19	0.36	1.14	0.46
Logistic (devol. ret. scale)	4.70	1.14	4.73	1.32	4.38	1.59
CNN1D (image scale)	7.20	1.66	7.88	1.84	7.85	2.61
CNN1D (cum. ret. scale)	5.33	1.71	5.36	1.52	5.45	1.85
CNN1D (devol. ret. scale)	-0.13	0.07	2.03	0.34	1.25	0.16
20-Day Return Horizon						
CNN	2.35	0.45	2.16	0.49	1.29	0.17
Logistic (image scale)	2.07	0.66	1.99	0.44	1.23	0.34
Logistic (cum. ret. scale)	0.51	0.11	0.41	0.33	0.43	0.30
Logistic (devol. ret. scale)	1.42	0.90	1.44	0.78	1.73	0.71
CNN1D (image scale)	2.49	0.70	2.72	0.59	2.45	0.88
CNN1D (cum. ret. scale)	1.47	0.67	1.45	0.57	1.46	0.44
CNN1D (devol. ret. scale)	-0.46	0.01	0.69	0.19	0.97	0.12
60-Day Return Horizon						
CNN	1.30	0.47	0.37	0.32	0.43	0.23
Logistic (image scale)	1.19	0.54	0.75	0.43	0.74	0.63
Logistic (cum. ret. scale)	-0.04	-0.07	0.07	0.19	0.13	0.17
Logistic (devol. ret. scale)	0.23	0.29	0.27	0.30	0.38	0.13
CNN1D (image scale)	1.28	0.48	1.04	0.44	0.76	0.49
CNN1D (cum. ret. scale)	0.24	0.19	0.35	0.21	0.24	-0.06
CNN1D (devol. ret. scale)	0.19	0.00	0.20	-0.12	0.32	-0.06

Table IX

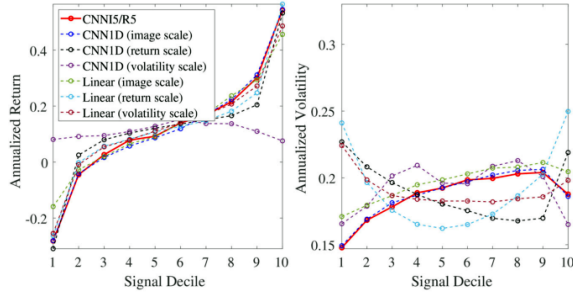


Figure 7. Prediction accuracy of CNN and logistic models by decile. The left panel plot average realized returns in each decile of I5/R5 CNN model forecasts, as well as for each decile forecasts from 1D CNN models and logistic models. Return averages are based on cross-sectional averages each period that are then averaged over time. The right panel plots the time-series volatility of decile returns. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 7

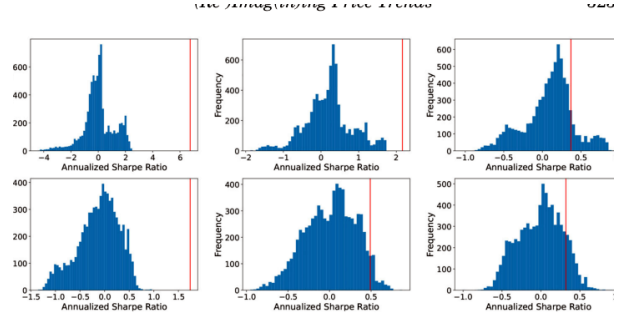


Figure 8. Comparison with traditional technical indicators. The figure shows histogram Sharpe ratios for 7,846 technical analysis portfolios that rebalance by week, month, and quarter, respectively. The top panels show equal-weight portfolios and the bottom panels show value-weight portfolios. Red vertical bars are the Sharpe ratios for the I20/R5, I20/R20, and I20/R40 NN strategies, respectively. (Color figure can be viewed at wileyonlinelibrary.com)

(b) Figure 8

6.8 Table X and Figure 9: International transfer and H-L decile portfolio Sharpe ratios

Read: U.S.-trained CNNs perform well in foreign markets, especially smaller markets with less local training data.

Interpretation: Transfer learning is strong external validation: the patterns are not purely U.S.-sample artifacts.

Economic message: Price patterns appear partially global, but large markets may benefit from local retraining.

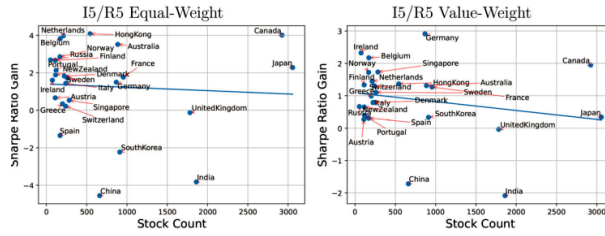


Figure 9. Sharpe ratio gains from international transfer. The panels show annualized out-of-sample Sharpe ratio gains from I5/R5 transfer learning (both equal-weight and value-weight strategies) versus the average number of stocks in each country. (Color figure can be viewed at wileyonlinelibrary.com)

(a) Figure 9

Table X
International Transfer and H-L Decile Portfolio Sharpe Ratios (I5/R5)
This table reports annualized out-of-sample Sharpe ratios for H-L decile spread portfolios within each country. We report the average monthly stock count by country, the image-based strategy that trains the I5/R5 CNN using local data, and the image-based strategy that directly transfers the I5/R5 model estimated in U.S. data without retraining. ***, **, and * indicate significant at the 1%, 5%, and 10% significance level, respectively.

	Stock Count	Equal-Weight			Value-Weight		
		Retrain	Direct Transfer	Transfer–Retrain	Retrain	Direct Transfer	Transfer–Retrain
U.S.	7,298	7.15	5.73	1.49	0.46	–2.68	–3.14
Global	17,296	3.056	3.56	5.84	2.38***	0.96	1.30
Japan	2,924	9.01	10.02	4.01***	2.98	4.93	1.95***
Canada	1,861	2.52	–1.30	–3.82	0.67	–1.42	–2.09
India	1,793	0.03	–0.10	–0.12	1.94	1.01	–0.09
United Kingdom	955	2.47	4.23	1.77***	1.12	2.39	1.27***
France	911	3.64	1.41	–2.23	1.74	2.07	0.33*
South Korea	886	8.28	11.79	3.52***	2.78	4.09	1.31***
Australia	888	–0.29	1.21	1.50***	–0.01	2.91	2.91***
Germany	236	2.36	–2.39	–4.55	0.66	–1.05	–1.72
Hong Kong	543	1.97	6.05	4.08***	0.72	2.09	1.37***
Singapore	284	6.98	7.51	0.53**	2.48	4.21	1.73***
Sweden	280	5.43	7.15	1.72***	0.83	1.93	1.10***
Italy	241	2.14	3.57	1.43***	0.76	1.56	0.79***
Switzerland	240	0.48	0.79	0.21	1.39	2.61	1.21***
Denmark	223	1.94	3.78	1.82***	1.18	1.96	0.79***
Netherlands	212	–0.20	3.68	3.88***	0.11	1.56	1.45***
Greece	201	2.74	3.08	0.34*	0.98	1.97	0.99***
Belgium	171	0.73	4.55	3.82***	0.73	2.91	2.17***
Spain	170	1.62	0.28	–1.34	0.68	0.99	0.31*
Norway	169	0.79	3.64	2.85***	1.11	2.83	1.72***
Portugal	121	0.30	2.43	2.13***	0.93	1.31	0.38**
New Zealand	114	0.50	2.38	1.88***	0.65	1.39	0.65***
Finland	113	2.66	5.33	2.66***	0.95	2.28	1.33***
Austria	110	0.14	0.80	0.66***	0.66	0.94	0.27
Ireland	75	0.47	2.07	1.60***	0.31	2.63	2.32***
Russia	53	–0.72	1.97	2.69***	–0.12	0.53	0.65***
Average	1,274	2.21	3.65	1.44	0.99	1.75	0.76
Average (excluding Global)	661	2.28	3.57	1.28	1.01	1.92	0.91

(b) Table X

6.9 Tables XI–XII: Time-scale transfer

Table XI
Time-Scale Transfer I5/R5 to I20/R20

This table reports results on the performance of strategies using time-scale transfer learning. In particular, we directly apply the I5/R5 CNN model estimated from daily data to construct forecasts from images that include 20 days of data sampled once every four days. Transfer learning portfolio performance (with a 20-day holding-period) is reported under “Transfer”. For comparison, we report a “Baseline” strategy that uses the I20/R20 CNN model of Tables II and IA, III, and a CNN model that is trained from scratch using images of 20 days of data sampled once every four days under “Retrain”. Finally, “Baseline+Transfer” shows the performance of an equal-weight average of the baseline and transfer strategies. ***, **, and * indicate significant at the 1%, 5%, and 10% significance level, respectively.

	Panel A: Equal-Weight Portfolios							
	Baseline		Retrain		Transfer		Baseline+Transfer	
	Ret	SR	Ret	SR	Ret	SR	Ret	SR
Low	–0.02	–0.12	–0.02	–0.10	–0.01	–0.07	–0.04	–0.19
2	0.05	0.28	0.05	0.28	0.04	0.25	0.04	0.20
3	0.07	0.39	0.07	0.37	0.07	0.40	0.06	0.35
4	0.09	0.49	0.09	0.49	0.08	0.43	0.08	0.44
5	0.11	0.59	0.10	0.56	0.09	0.50	0.09	0.52
6	0.11	0.59	0.11	0.61	0.10	0.53	0.11	0.64
7	0.13	0.74	0.12	0.65	0.11	0.59	0.13	0.70
8	0.14	0.81	0.14	0.79	0.14	0.70	0.15	0.80
9	0.15	0.87	0.15	0.82	0.17	0.84	0.17	0.89
High	0.18	1.04	0.20	1.15	0.24	1.14	0.23	1.17
H-L	0.21***	1.73*	0.22***	1.77*	0.25***	1.76*	0.26***	1.75*
Turnover								

(Continued)

Table XI—Continued

	Panel B: Value-Weight Portfolios							
	Baseline		Retrain		Transfer		Baseline+Transfer	
	Ret	SR	Ret	SR	Ret	SR	Ret	SR
Low	0.05	0.28	0.06	0.38	0.03	0.22	0.04	0.24
2	0.04	0.21	0.06	0.36	0.05	0.34	0.05	0.32
3	0.05	0.32	0.05	0.34	0.06	0.42	0.04	0.27
4	0.05	0.33	0.08	0.53	0.05	0.35	0.05	0.30
5	0.06	0.40	0.07	0.42	0.07	0.46	0.06	0.40
6	0.08	0.50	0.08	0.50	0.08	0.50	0.07	0.45
7	0.08	0.51	0.08	0.54	0.09	0.56	0.07	0.45
8	0.09	0.58	0.09	0.58	0.11	0.66	0.10	0.61
9	0.08	0.55	0.10	0.64	0.12	0.65	0.10	0.61
High	0.11	0.67	0.10	0.66	0.12	0.69	0.12	0.73
H-L	0.05**	0.40	0.04	0.30	0.09***	0.73	0.08***	0.67
Turnover								

Read: Table XI transfers I5/R5 to I20/R20, and Table XII transfers I5/R5 to I60/R60. Transfer performs well and combines with baseline forecasts.

Interpretation: Short-horizon learned patterns can remain useful at lower frequencies.

Economic message: This suggests self-similarity in price patterns and motivates new nonlinear theories of price dynamics.

7 Interpretive synthesis

- **Representation matters.** Images preserve joint shape information that standard returns summaries discard.
- **Model flexibility matters.** CNN filters can detect nonlinear local configurations.
- **Out-of-sample validation matters.** Fixed training and long testing discipline the machine-learning exercise.
- **External validity matters.** Transfer learning shows the signal travels across countries and time scales.

Table XII
Time-Scale Transfer 15/R5 to 160/R60

This table reports results on the performance of strategies using time-scale transfer learning. In particular, we directly apply the 15/R5 CNN model estimated from daily data to construct forecasts from images that include 60 days of data sampled once every 12 days. Transfer-learning portfolio performance (with a 60-day holding period) is reported under "Transfer." For comparison, we report a "Baseline" strategy that uses the 160/R60 CNN model of Tables II and IA-III, and a CNN model that is trained from scratch using images of 60 days of data sampled once every 12 days under "Retrain". Finally, "Baseline+Transfer" shows the performance of an equal-weight average of the baseline and transfer strategies. ***, **, and * indicate significant at the 1%, 5%, and 10% significance level, respectively.

	Panel A: Equal-Weight Portfolios							
	Baseline		Retrain		Transfer		Baseline+Transfer	
	Ret	SR	Ret	SR	Ret	SR	Ret	SR
Low	0.08	0.27	0.08	0.30	0.06	0.31	0.06	0.22
2	0.09	0.37	0.08	0.34	0.08	0.41	0.09	0.35
3	0.11	0.46	0.10	0.43	0.10	0.50	0.11	0.46
4	0.12	0.52	0.12	0.53	0.11	0.54	0.11	0.51
5	0.12	0.55	0.12	0.55	0.11	0.54	0.12	0.55
6	0.12	0.57	0.12	0.57	0.13	0.61	0.12	0.61
7	0.14	0.68	0.13	0.64	0.12	0.55	0.14	0.67
8	0.14	0.72	0.13	0.64	0.14	0.60	0.14	0.69
9	0.14	0.75	0.13	0.72	0.14	0.59	0.15	0.78
High	0.15	0.88	0.14	0.88	0.17	0.69	0.16	0.93
H-L	0.07*		0.06		0.10***		0.10***	
Turnover	58%	0.43	59%	0.37	59%	0.90	58%	0.81

(Continued)

Table XII—Continued

	Panel B: Value-Weight Portfolios							
	Baseline		Retrain		Transfer		Baseline+Transfer	
	Ret	SR	Ret	SR	Ret	SR	Ret	SR
Low	0.07	0.32	0.11	0.50	0.06	0.38	0.08	0.37
2	0.10	0.50	0.08	0.41	0.07	0.34	0.08	0.40
3	0.08	0.46	0.07	0.39	0.08	0.47	0.08	0.41
4	0.09	0.47	0.06	0.35	0.09	0.58	0.08	0.46
5	0.08	0.43	0.09	0.50	0.10	0.61	0.09	0.49
6	0.09	0.49	0.08	0.48	0.09	0.55	0.08	0.45
7	0.09	0.54	0.10	0.55	0.09	0.53	0.09	0.49
8	0.09	0.53	0.10	0.58	0.10	0.56	0.10	0.57
9	0.09	0.52	0.09	0.55	0.11	0.65	0.09	0.58
High	0.11	0.75	0.11	0.77	0.10	0.51	0.11	0.81
H-L	0.03	0.23	0.00	0.01	0.04	0.27	0.03	0.31
Turnover	59%		61%		62%		59%	

8 Short summary

- The paper applies CNN image analysis to stock price charts.
- It uses CRSP daily OHLC and volume data from 1993–2019.
- Images summarize 5, 20, or 60 days of market history and forecast 5, 20, or 60-day return signs.
- CNN signals outperform standard technical indicators in out-of-sample portfolio tests.
- Performance survives longer horizons, large-stock restrictions, and diagnostic controls.
- International and time-scale transfer show that learned price patterns are not purely sample-specific.

9 Conclusion

9.1 Core conclusion

Jiang, Kelly, and Xiu show that price charts contain predictive information that is difficult to capture with standard hand-crafted trend signals.

9.2 Economic intuition

Historical prices contain nonlinear shapes and interactions among direction, volatility, moving averages, and volume. CNNs can learn these shapes directly.

9.3 Takeaway

The paper reopens technical analysis as a disciplined empirical asset-pricing topic: visual representation plus machine learning reveals price-trend predictability that standard variables miss.